RAPID COMMUNICATION | AUGUST 30 2011

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J. Chem. Phys. 135, 081106 (2011)
https://doi.org/10.1063/1.3633329

Communication: Extended multi-state complete active space second-order perturbation theory: Energy and nuclear gradients

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(Received 30 July 2011; accepted 16 August 2011; published online 30 August 2011)

The extended multireference quasi-degenerate perturbation theory, proposed by Granovsky [J. Chem. Phys. **134**, 214113 (2011)], is combined with internally contracted multi-state complete active space second-order perturbation theory (XMS-CASPT2). The first-order wavefunction is expanded in terms of the union of internally contracted basis functions generated from all the reference functions, which guarantees invariance of the theory with respect to unitary rotations of the reference functions. The method yields improved potentials in the vicinity of avoided crossings and conical intersections. The theory for computing nuclear energy gradients for MS-CASPT2 and XMS-CASPT2 is also presented and the first implementation of these gradient methods is reported. A number of illustrative applications of the new methods are presented. © *2011 American Institute of Physics*. [doi:10.1063/1.3633329]

Despite the remarkable success of single-reference many-body perturbation and coupled-cluster theories for predictive simulations of molecular electronic structures, there are yet many systems that require multi-reference (MR) approaches: this is particularly the case for systems with quasi-degenerate electronic states near avoided crossings or conical intersections. Such states are often important in photochemical relaxation processes. Another example is the computation of many-electron excited states of molecules, for which one would need to include high-rank excitations in single-reference linear response methods.

Second-order multireference perturbation theories (MRPT2) have been widely used as efficient approaches for treating dynamical electron correlation. Among the many proposed variants of MRPT2 (Refs. 1–16) the complete active space second-order perturbation (CASPT2) methods[1–6] are particularly popular. The CASPT2 method uses internally contracted configuration spaces, which exactly span the first-order interaction space of the reference wavefunction. This has the advantage that the number of parameters in the first-order wavefunction is independent of the number of reference configurations, and therefore very long complete active space self-consistent field (CASSCF) configuration expansions can be used as reference wavefunctions. However, since the internally contracted configurations are state-specific, errors may arise in near degeneracy situations. This problem can be overcome by multi-state (MS) CASPT2,[17] in which the wavefunctions of different states are allowed to mix. The mixing coefficients are obtained by diagonalization of an effective Hamiltonian. It should be mentioned that prior to MS-CASPT2 Nakano has proposed a similar method, called multiconfiguration quasi-degenerate perturbation theory (MCQDPT),[18] using uncontracted configuration state functions (CSFs).

Recently, Granovsky[19] showed that the conventional MCQDPT and MS-CASPT2 may lead to singular potential energy surfaces around avoided crossings and conical intersections. This is ascribed to the fact that the zeroth-order energies that come into the Hylleraas functional are not smoothly changing, even though the sum of zeroth- and first-order energies (i.e., CASSCF energies) are. In order to ameliorate this problem, he has proposed a new family of methods called "extended" multiconfiguration quasi-degenerate perturbation theories (XMCQDPT), which employ the entire block of the Fock operator in the reference space in the zeroth-order Hamiltonian.[19] Although the formal theory proposed by Granovsky is quite general, the implementation in the FIREFLY package[20] is limited to the MCQDPT2 with uncontracted CSFs.

Analytical energy gradients with respect to the nuclear coordinates are of utmost importance for geometry optimizations and for *ab initio* dynamics. For example, the availability of CASPT2 energy gradients makes it possible to optimize geometries of states with strong multi-reference character, which occur, for example, in many transition metal complexes that are of interest in catalysis, or in electronically excited states. However, since the CASPT2 wavefunctions depend on the CASSCF orbitals (possibly state-averaged), and the internally contracted configurations depend on the coefficients of the reference wavefunction, the theory for CASPT2 gradients is quite involved, and the first CASPT2 gradient program was presented only in 2003.[21] Gradients for multi-state MRPT2 have also been implemented by Nakano and co-workers,[22,23] as well as for a generalized Van Vleck perturbation theory[24] by Hoffmann and co-workers.[25] Although the gradient code for the MS-CASPT2 method has been available in MOLPRO for some years[26,27] and used by a number of authors,[28–30] its theory has not been described in the literature.

In this work, we generalize the XMCQDPT method for internally contracted MS-CASPT2 and define the XMS-CASPT2 method. Moreover, we formulate the theory for

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TABLE I. Definition of orbital and configuration labels.

i, j, k, l	any occupied orbitals in CASSCF
a, b	external orbitals (unoccupied in CASSCF)
r, s	any orbitals
R	reference configurations
I, J	internal CSFs (no electrons in virtual orbitals)
S^a	singly external CSFs
n	states included in CASSCF
M, N	states included in CASPT2

analytic MS-CASPT2 and XMS-CASPT2 nuclear energy gradients, and report their implementation in the MOLPRO[26,27] program. We will only outline the essential parts of the theory in this communication. The details and all working equations can be found in the supplementary material.[31]

The index notation used in this paper is summarized in Table I. The zeroth-order Hamiltonian of the XMS-CASPT2 method is defined as[19]

$$\hat{H}^{(0)} = \hat{P}\hat{f}\hat{P} + \hat{Q}\hat{f}\hat{Q}, \tag{1}$$

where $\hat{P}$ is the projector onto the reference space, i.e., $\hat{P} = \sum_N |N\rangle\langle N|$ with $|N\rangle = \sum_R c_N^R |R\rangle$ being the Nth reference function, and $\hat{Q}$ is its complementary projector. Note that the XMS-CASPT2 reference states $|N\rangle$ are in general a subset of the states included in the state-averaged CASSCF, and therefore different labels (N and n, respectively) are used in the two cases. The CASPT2 Fock operator $\hat{f}$ and its representation f_{rs} in the molecular orbital basis are defined as

$$\hat{f} = \sum_{rs} f_{rs}\hat{E}_{rs}, \tag{2}$$

$$f_{rs} = h_{rs} + \sum_{ij} d_{ij}\left(J_{rs}^{ij} - \frac{1}{2}K_{rs}^{ij}\right), \tag{3}$$

where h_{rs} are the one-electrons integrals in the molecular Hamiltonian, and $J_{rs}^{ij} = (ij|rs)$ and $K_{rs}^{ij} = (ir|js)$ are two-electron integrals (in Mulliken notation). d_{ij} are elements of the state-averaged one-particle reduced density matrix in CASPT2. We assume equal weights for all states included in the XMS-CASPT2, which ensures the invariance of the Fock operator with respect to rotations among the reference states.

In general, the Fock operator is not diagonal in the space of reference wavefunctions,

$$f_{MN} = \langle M|\hat{f}|N\rangle = \sum_{R,R'}\sum_{ij} f_{ij} c_M^R \langle R|\hat{E}_{ij}|R'\rangle c_N^{R'}. \tag{4}$$

The difference between the zeroth-order Hamiltonians of the XMS-CASPT2 and conventional MS-CASPT2 methods lies in the first term of Eq. (1): MS-CASPT2 only contains the diagonal terms $|N\rangle f_{NN}\langle N|$, while XMS-CASPT2 also includes the off-diagonal terms $|M\rangle f_{MN}\langle N|$ (with $M \neq N$). Using the unitary matrix U_{MN} that satisfies

$$\mathbf{U}^\dagger \mathbf{f} \mathbf{U} = \tilde{\mathbf{f}}, \tag{5}$$

with $\tilde{\mathbf{f}}$ being a diagonal matrix, we define the rotated reference functions $|\tilde{N}\rangle = \sum_M |M\rangle U_{MN}$. In this basis the off-diagonal Fock matrix elements are zero. The zeroth-order Hamiltonian can then be rewritten as

$$\hat{H}^{(0)} = \sum_N |\tilde{N}\rangle \tilde{f}_{NN}\langle\tilde{N}| + \hat{Q}\hat{f}\hat{Q}, \tag{6}$$

which has similar structure to the zeroth-order Hamiltonian of the conventional MS-CASPT2 method, apart from the use of the rotated references $|\tilde{N}\rangle$ rather than $|N\rangle$.

The first-order wavefunction is parameterized as

$$|\tilde{\Psi}_N\rangle = |\tilde{N}\rangle + |\Psi_N^{(1)}\rangle, \tag{7}$$

$$|\Psi_N^{(1)}\rangle = \sum_I t_N^I |I\rangle + \sum_{S,a} t_{a,N}^S |S^a\rangle + \frac{1}{2}\sum_{ij,ab,M} t_{ab,NM}^{ij} |\Phi_{ij,M}^{ab}\rangle, \tag{8}$$

where $|\Phi_{ij,M}^{ab}\rangle$ are internally contracted doubly external configurations generated from the Mth rotated reference function, i.e., $|\Phi_{ij,M}^{ab}\rangle = \hat{E}_{ij}^{ab}|\tilde{M}\rangle$. The doubly external part of the first-order wavefunctions [i.e., the last term of Eq. (8)] is expanded by the union of internally contracted basis functions from all the reference functions (the so-called MS-MR-CASPT2 scheme). This ensures invariance of the expansion space with respect to unitary rotations of the reference functions. A similar ansatz has been used earlier by Knowles and Werner[32] for internally contracted multi-state MRCI. As pointed out by Granovsky,[19] this invariance property allows for the efficient implementation by diagonalizing the zeroth-order Hamiltonian in the reference space. This cannot be achieved by the original MS-CASPT2 formulation of Finley *et al.*,[17] in which the expansion space is state specific. We note that in principle the internal and singly external configurations could also be internally contracted, but in the current work this has not been done in order to avoid higher-order reduced density matrices and to simplify the analytical energy gradients. Perhaps most efficient would be the partially contracted scheme as proposed by Celani and Werner for CASPT2.[5] Recently, this has also been used for MRCI,[33] and an implementation for XMS-CASPT2 will be done in the near future.

To determine the first-order wavefunctions, we first solve the amplitude equation for each (rotated) reference function,

$$\langle\Omega|\hat{H}^{(0)} - E_N^{(0)} + E_{\mathrm{shift}}|\Psi_N^{(1)}\rangle + \langle\Omega|\hat{H}|\tilde{N}\rangle = 0, \tag{9}$$

where $|\Omega\rangle$ spans the projection manifold consisting of $|I\rangle, |S^a\rangle, |\Phi_{ij,M}^{ab}\rangle$ and E_{shift} is the level shift parameter.[34] Note that the projection to the internal space excludes the reference configurations. Using the amplitudes determined in this way, the effective Hamiltonian up to second-order is formed as

$$(H_{\mathrm{eff}})_{MN} = \tfrac{1}{2}(\langle\tilde{\Psi}_M|\hat{H}|\tilde{N}\rangle + \langle\tilde{M}|\hat{H}|\tilde{\Psi}_N\rangle) - \delta_{MN}E_{\mathrm{shift}}\langle\Psi_M^{(1)}|\Psi_M^{(1)}\rangle, \tag{10}$$

in which we employ the symmetric form adopted by Nakano.[18] It should be noted that the level shift slightly breaks the invariance with respect to the rotation of reference functions. Diagonalization of this effective Hamiltonian gives the

XMS-CASPT2 energies. The resulting wavefunctions are

$$|\Psi_N\rangle = \sum_M |\tilde{\Psi}_M\rangle T_{MN}, \tag{11}$$

where $\mathbf{T}$ are the normalized eigenvectors of $\mathbf{H}_{\text{eff}}$. From the observation that these equations are exactly the same as those of the conventional MS-CASPT2, except that the rotated reference functions are used,[19] it is clear that one can implement the computer codes for XMS-CASPT2 energies by slightly modifying the conventional MS-CASPT2 program.[35]

Our nuclear gradient theory is formulated by the Lagrangian approach. If the Lagrangian is made stationary with respect to all parameters, the gradient is obtained by evaluating the Lagrangian with derivative AO integrals. The Lagrangian of the XMS-CASPT2 method for the Nth state reads

$$\begin{aligned} L_N = {} & (\mathbf{T}^\dagger \mathbf{H}_{\text{eff}} \mathbf{T})_{NN} - \eta[(\mathbf{T}^\dagger\mathbf{T})_{NN} - 1] + \sum_{M \neq M'} w_{MM'} \tilde{f}_{MM'} \\ & + \sum_M \sum_\Omega \lambda_{\Omega,MN} \big[\langle \Omega | \hat{f} - E_M^{(0)} + E_{\text{shift}} | \tilde{\Psi}_M \rangle \\ & + \langle \Omega | \hat{H} | \tilde{M} \rangle \big] + \frac{1}{2}\text{tr}[\mathbf{Z}^\dagger(\mathbf{A} - \mathbf{A}^\dagger)] - \frac{1}{2}\text{tr}[\mathbf{X}(\mathbf{C}^\dagger\mathbf{S}\mathbf{C} - 1)] \\ & + \sum_n W_n \big[\mathbf{z}_n^\dagger \big(\mathbf{H} - E_n^{\text{ref}}\big)\mathbf{c}_n - \tfrac{1}{2} x_n \big(\mathbf{c}_n^\dagger \mathbf{c}_n - 1\big) \big] \\ & + \sum_i^{\text{core}} \sum_j^{\text{closed}} z_{ij} f_{ij}^{\text{ref}}. \end{aligned} \tag{12}$$

The first term is the XMS-CASPT2 energy expression. The following terms in the first two lines of Eq. (12) account for the XMS-CASPT2 variational conditions. Among these, the terms $w_{MM'}\tilde{f}_{MM'}$ are specific to the XMS-CASPT2 method. They account for the diagonalization of the Fock operator in the reference space [Eq. (5)]. The remaining terms in the last three lines of Eq. (12) arise from the variational conditions of the CASSCF reference function. $\mathbf{A} - \mathbf{A}^\dagger$ is the orbital gradient of CASSCF, which is multiplied by Lagrange multipliers $\mathbf{Z}$ (the explicit formulas are given in the supplementary material). The next term accounts for the orthonormality condition of the orbitals, where $\mathbf{C}$ is the molecular orbital coefficient matrix and $\mathbf{S}$ is the overlap matrix of atomic orbitals. The second last line of Eq. (12) accounts for the conditions to determine the CASSCF reference coefficients $\mathbf{c}_n$. E_n^{ref} is the CASSCF energy for the nth state, W_n are the weights in the state-averaged energy expression, and $(\mathbf{H})_{IJ} = \langle I|\hat{H}|J\rangle$. The convention $\mathbf{z}_n^\dagger \mathbf{c}_n = 0$ is used as in Ref. 21. The last term accounts for the condition that the inactive (doubly occupied) orbitals must diagonalize the CASSCF effective Fock matrix $\mathbf{f}^{\text{ref}}$. This is required since the CASPT2 energy depends on rotations between (uncorrelated) core orbitals i and correlated closed-shell orbitals j. Note that the Fock matrix f_{ij}^{ref} is not necessarily the same as f_{ij} used in CASPT2, since additional states may be included in the CASSCF calculation.

The computation proceeds as follows: First we obtain $\lambda_{\Omega,NM}$ by solving the so-called λ-equation, which is derived by taking the derivative of L_N with respect to t_M^I, $t_{a,M}^S$, and $t_{ab,MM'}^{ij}$. We then compute $w_{MM'}$ by solving the equation that is obtained by the stationary condition of L_N with respect to the rotation of the reference functions U. Given $\lambda_{\Omega,MN}$ and $w_{MM'}$, we solve the Z vector equations for the orbital rotations ($\mathbf{Z}$, $\mathbf{X}$, and z_{ij}) and the contraction coefficients ($\mathbf{z}_n$ and x_n) as in the standard (MS) CASPT2 gradient theory.[21] Finally, effective one and two particle density matrices are formed, transformed into the AO basis, and multiplied by the derivative one- and two-electron integrals, respectively. All the working equations are explicitly shown in the supplementary material.

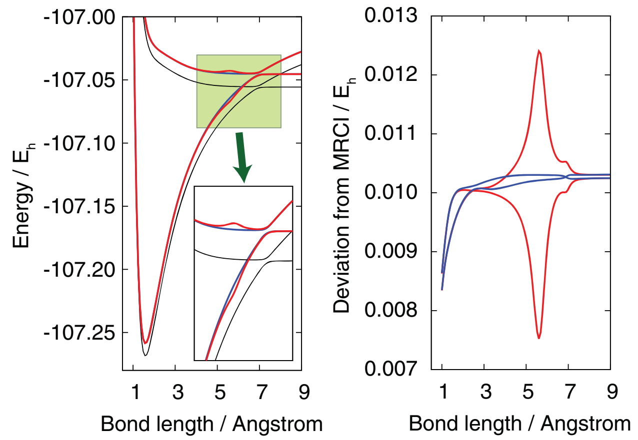

FIG. 1. LiF potential energy curves computed by internally contracted MS- and XMS-CASPT2 (left panel). The difference from MRCI is shown in the right panel. Red curves: MS-CASPT2; blue curves: XMS-CASPT2; black curves: MRCI.

The covalent–ionic avoided crossing of LiF at a long distance has been a notorious test ground for quasi-degenerate perturbation theories (e.g., Refs. 17–19 and 36). Here we have used an active space consisting of six electrons in three a_1, two b_2, and two b_1 orbitals with the frozen core approximation. Dunning's aug-cc-pVTZ basis set has been used.[37,38] For comparison, the MRCI (Refs. 39 and 40) curves (without Davidson's correction, see the detailed discussion by Varandas[41]) have been computed. They are shown in Fig. 1. The MS-CASPT2 curve has a large hump of about 3 mE_h around 5.3 Å. This artificial hump does not appear in the XMS-CASPT2 curves; in fact, in the long-range part of the potential the curves are almost parallel to the MRCI ones. The superior performance of XMS-CASPT2 to MS-CASPT2 is obvious in this example.

Two of the authors have previously studied the low-lying exited states of Pyrrole using state-specific CASPT2, and found that there is strong mixing in the first and second excited Rydberg states in $^1B_1$ symmetry.[21] Here we compute the vertical and adiabatic excitation energies of the lowest two states in the $^1B_1$ and $^1A_2$ symmetries by the standard MS-CASPT2 and the newly developed XMS-CASPT2 methods. The vertical excitation energies are compared with more accurate MRCI+Q (Refs. 39 and 40) and CC3 (Ref. 42) calculations.

The results are presented in Table II. The computational details follow those in Ref. 21. Dunning's aug-cc-pVTZ basis set[37,38] is used without d functions for H. Only π electrons are active and the active orbitals consist of the $10-11a_1$, $1-3b_1$, $7b_2$, and $1-2a_2$ orbitals. In CASSCF, $1\,^1A_1$, $1-2\,^1B_1$,

TABLE II. Vertical and adiabatic excitation energies for low-lying excited states of pyrrole in eV.

Method	1^1B_1 $2b_1 \to 3s$	2^1B_1 $1a_2 \to 3p_y$	1^1A_2 $1a_2 \to 3s$	2^1A_2 $1a_2 \to 3p_z$
	Vertical excitation			
CASPT2	5.91	5.99	5.21	6.00
MS-CASPT2	5.77	6.07	5.19	6.00
XMS-CASPT2	5.87	5.98	5.20	5.99
MRCI+Q[a]	5.83	5.95	5.08	5.86
CC3[b]	5.85	6.00	5.10	5.86
	Adiabatic excitation			
MS-CASPT2	5.72	5.99	4.96	5.84
XMS-CASPT2	5.73	5.94	4.97	5.83

[a]Computed at the CASPT2 ground-state geometry.
[b]Taken from Ref. 43.

and $1-2\,^1A_2$ states are averaged. The level shift[34] of 0.2 E_h was used to avoid intruder state problems. Using the new analytic gradient code, geometries have been optimized (within C_{2v}) for each method and state except for the MRCI+Q that used the XMS-CASPT2 ground state geometry. We note that the level shift is now treated exactly in the gradient code, which was not the case in Ref. 21. All CASPT2 and MRCI+Q calculations were performed with MOLPRO.[26,27] The CC3 results, which were computed with a slightly different basis set and geometry, were taken from Ref. 43.

The XMS-CASPT2 vertical excitation energies to the nearly degenerate 1^1B_1 ($2b_1 \to 3s$) and 2^1B_1 ($1a_2 \to 3p_y$) Rydberg states show better agreement with those computed by MRCI+Q and CC3 as compared to the standard MS-CASPT2 ones. The MS-CASPT2 and XMS-CASPT2 wavefunctions were rotated from the CASSCF reference functions by 37.3° and 21.0°, respectively. The standard MS-CASPT2 method overestimates the mixing of the states, which lowers the 1^1B_1 energy by about 0.1 eV and rises the 2^1B_1 energy by a similar amount. The adiabatic excitation energies are less affected, but the 2^1B_1 excitation energies still differ by 0.05 eV. In contrast, the MS-CASPT2 and XMS-CASPT2 methods give almost identical results for the A_2 Rydberg states. The optimized geometrical parameters are presented in the supplementary material.

In conclusion, we have proposed and implemented the XMS-CASPT2 method that rectifies the shortcomings of the standard MS-CASPT2 method: the overestimation of off-diagonal elements of the effective Hamiltonian when state mixing is strong. The definition of the zeroth-order Hamiltonian is taken from the preceding work of Granovsky on XMCQDPT,[19] with which we have combined internal contraction used in CASPT2 to allow for application to systems which require large active spaces. The analytic nuclear gradients for the MS-CASPT2 and XMS-CASPT2 methods have also been implemented. The computer code for these methods will be made available in the next release of the MOLPRO package.[27]

Dr. Gerald Knizia is thanked for helpful discussions. This work has been supported by the Deutsche Forschungsgemeinschaft (DFG) as part of the SimTech Cluster of Excellence at the University of Stuttgart. T.S. is supported by the Japan Society for the Promotion of Science Research Fellowship for Research Abroad.

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